FIVE EQUIVALENT DESCRIPTIONS OF QUASI-PROJECTIVE MORPHISMS OVER A BASE WITH AN AMPLE INVERTIBLE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13839-13854 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13856-13889, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Hub lemma: several later chapter proofs cite this equivalence; collected for the explicit (3) implies (4)/(5) construction.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The Morphisms chapter dictionary between quasi-projective, H-quasi-projective, open-H-projective and very-ample descriptions (lemmas H-quasi-projective-quasi-projective, projective-over-quasi-projective-is-H-projective, H-quasi-projective-open-H-projective, quasi-projective-finite-type-over-S, ample-very-ample, pullback-ample-tensor-relatively-ample, finite-type-ample-very-ample).

Source reference. Original lemma 01WA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VU: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0892: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13889–13889. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme which has an ample invertible sheaf. Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *$X \to S$ is quasi-projective,*
(2) *$X \to S$ is H-quasi-projective,*
(3) *there exists a quasi-compact open immersion $X \to X'$ of schemes over S with $X' \to S$ projective,*
(4) *$X \to S$ is of finite type and X has an ample invertible sheaf, and*
(5) *$X \to S$ is of finite type and there exists an f-very ample invertible sheaf.*

Proof. The implication (2) $\Rightarrow$ (1) is Morphisms, Lemma 01VY. The implication (1) $\Rightarrow$ (2) is Morphisms, Lemma 087S. The implication (2) $\Rightarrow$ (3) is Morphisms, Lemma 01WA

Assume $X \subset X'$ is as in (3). In particular $X \to S$ is of finite type. By Morphisms, Lemma 01WA the morphism $X \to S$ is H-projective. Thus there exists a quasi-compact immersion $i : X \to \mathbf{P}^n_S$. Hence $\mathcal{L} = i^*\mathcal{O}_{\mathbf{P}^n_S}(1)$ is f-very ample. As $X \to S$ is quasi-compact we conclude from Morphisms, Lemma 01VN that $\mathcal{L}$ is f-ample. Thus $X \to S$ is quasi-projective by definition.

The implication (4) $\Rightarrow$ (2) is Morphisms, Lemma 01VS.

The implication (5) $\Rightarrow$ (4) follows from Morphisms, Lemma 0892.

The equivalence of (1) and (5) follows from Morphisms, Lemmas 01VN and 01VU. □